# Adding and Subtracting Fractions with Like Denominators Answer Key

Grades 4–5

**Quick Answers**

1. $\frac{4}{5}$
2. $\frac{7}{8}$
3. $\frac{1}{3}$
4. $\frac{2}{3}$
5. $\frac{9}{8}$
6. $\frac{5}{9}$
7. $\frac{1}{2}$
8. $\frac{5}{6}$
9. $\frac{7}{8}$
10. $\frac{5}{7}$

**Curriculum**

**Level:** Grades 4–5
4.NF.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
*Difficulty 1–4 of 5 across 10 tagged checks.*

## 1. Fifths bar: $\frac{1}{5} + \frac{3}{5}$.

**Solution:** One fifth is 1 shaded part; three fifths is 3 more parts of the same size. On the bar that is $1 + 3 = 4$ shaded parts out of 5, so only the count of parts changes — the size of a part is still one fifth.

$$\frac{1}{5} + \frac{3}{5} = \frac{1+3}{5} = \boxed{\frac{4}{5}}$$

Already in simplest form (4 and 5 share no factor but 1).

## 2. Eighths bar: $\frac{2}{8} + \frac{5}{8}$.

**Solution:** Count eighths: 2 shaded, then 5 more shaded, giving $2 + 5 = 7$ eighths. The denominator stays 8 because the bar is still cut into 8 equal parts.

$$\frac{2}{8} + \frac{5}{8} = \frac{2+5}{8} = \boxed{\frac{7}{8}}$$

The bar is not quite full: 7 of the 8 parts are shaded.

**3. Ninths bar:** $\frac{7}{9} - \frac{4}{9}$.
**Solution:** Start with 7 shaded ninths and cross out 4 of them. $7 - 4 = 3$ ninths are still shaded.

$$\frac{7}{9} - \frac{4}{9} = \frac{7-4}{9} = \frac{3}{9}$$

Now simplify: 3 and 9 are both divisible by 3, and $3 \div 3 = 1$, $9 \div 3 = 3$.

$$\boxed{\frac{1}{3}}$$

On the bar, the 3 shaded ninths line up exactly with one third of the bar, which is why the two names describe the same amount.

**4. Sixths bar:** $\frac{5}{6} - \frac{1}{6}$.
**Solution:** $5 - 1 = 4$ sixths remain shaded.

$$\frac{5}{6} - \frac{1}{6} = \frac{4}{6}$$

Both 4 and 6 divide by 2, so $\frac{4}{6} = \boxed{\frac{2}{3}}$.

**5. Two eighths bars:** $\frac{5}{8} + \frac{4}{8}$.
**Solution:** Counting eighths across both bars: $5 + 4 = 9$ eighths.

$$\frac{5}{8} + \frac{4}{8} = \boxed{\frac{9}{8}}$$

Nine eighths is more than one whole. Eight of the nine eighths fill one whole bar and one eighth is left over, so the same amount is 1 whole and 1 eighth.
*Watch for:* an answer of $\frac{9}{16}$ — that comes from adding the denominators, which would mean the pieces got smaller when nothing was cut.

**6. Missing numerator:** $\frac{2}{9} + \_ = \frac{7}{9}$.
**Solution:** The pieces are all ninths, so the question is really "2 ninths plus how many ninths makes 7 ninths?" Count up on the bar from 2 to 7: that is $7 - 2 = 5$ more ninths.

$$\frac{2}{9} + \frac{5}{9} = \frac{7}{9} \quad \checkmark \qquad \boxed{\frac{5}{9}}$$

### 7. Chocolate bar story: $\frac{11}{12} - \frac{5}{12}$.

**Solution:** "Eats" means take away, so subtract. The bar is cut into 12 equal pieces, so both fractions count twelfths and the denominator does not change.

$$\frac{11}{12} - \frac{5}{12} = \frac{11-5}{12} = \frac{6}{12}$$

Simplify by dividing both parts by 6: $\boxed{\frac{1}{2}}$ of the bar is left.

*Watch for:* $\frac{16}{12}$ — that is what adding gives, and an answer larger than what Lena started with cannot be what is left.

### 8. Three shaders on one sixths bar: $\frac{1}{6} + \frac{2}{6} + \frac{2}{6}$.

**Solution:** All three friends shade sixths of the same bar, so add the numerators only: $1+2+2=5$.

$$\frac{1}{6} + \frac{2}{6} + \frac{2}{6} = \boxed{\frac{5}{6}}$$

The bar is *not* full: 5 of its 6 parts are shaded, so one sixth is still blank.

### 9. Work backwards: $\_ - \frac{3}{8} = \frac{4}{8}$.

**Solution:** The amount before pouring is the amount poured out plus the amount still there — undo the subtraction by adding.

$$\frac{3}{8} + \frac{4}{8} = \frac{7}{8}$$

Check by subtracting: $\frac{7}{8} - \frac{3}{8} = \frac{4}{8}$ ✓. The jug held $\boxed{\frac{7}{8}}$ of a jug before the pouring.

### 10. Find and fix: Jae wrote $\frac{3}{7} + \frac{2}{7} = \frac{5}{14}$.

**Solution:** Jae added the denominators as well as the numerators. That cannot be right: $\frac{5}{14}$ is smaller than $\frac{3}{7}$, and adding a positive amount can never make the total smaller. On a sevenths bar, shading 3 parts and then 2 more leaves 5 shaded parts, and the parts are still sevenths — nothing re-cut the bar into fourteenths.

$$\frac{3}{7} + \frac{2}{7} = \frac{3+2}{7} = \boxed{\frac{5}{7}}$$